EE313 Project Design Phase

Wideband amplifier with an automatic gain control

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"I affirm that I have not given or received any unauthorized help on this report and that this work is my own."

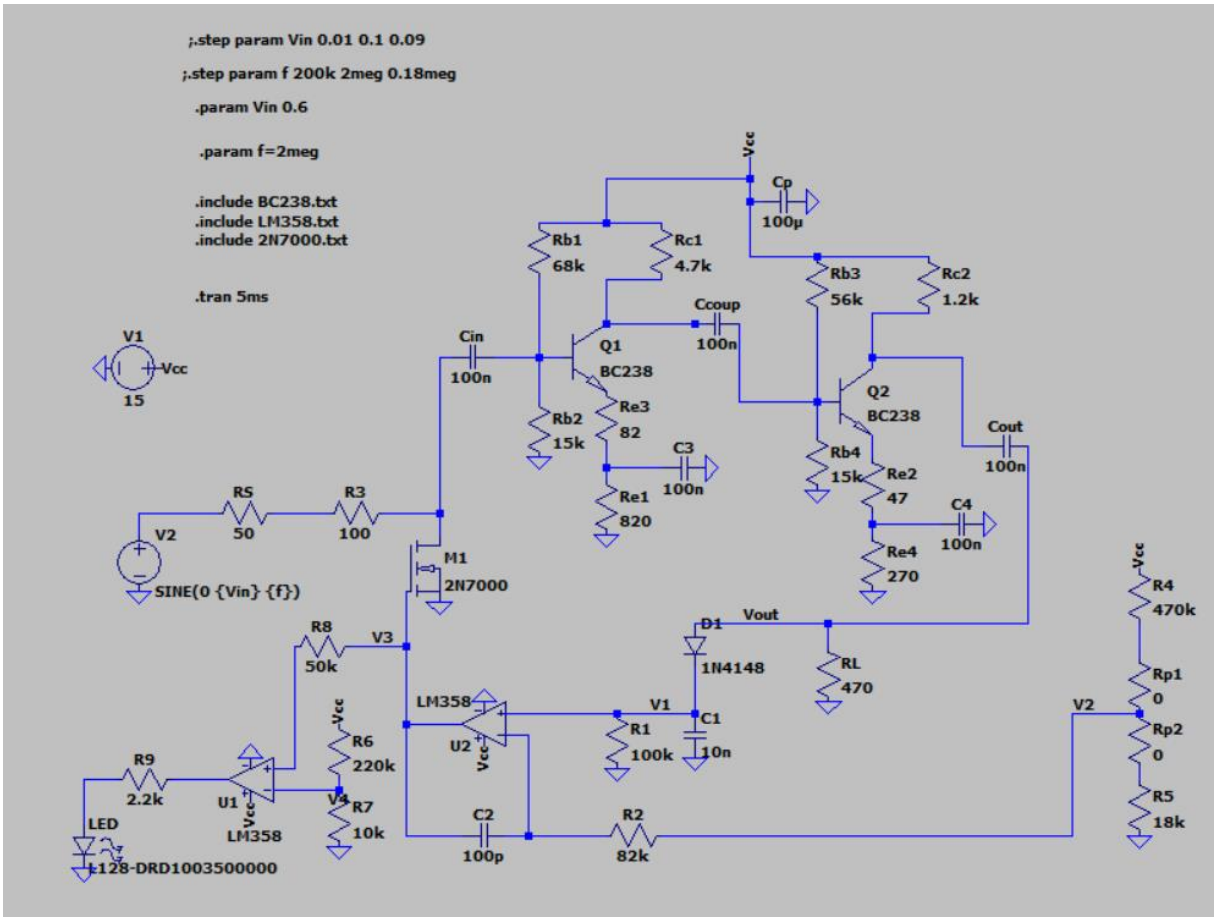

Figure 1: Photo of the circuit.

Specification 1:

The peak output voltage is Voutp=1±0.1 V undistorted sinusoidal when the peak input voltage is between Vinp=0.01 V and 0.1 V while the frequency is between 200KHz and 2MHz.

Input=200kHz and 0.01V

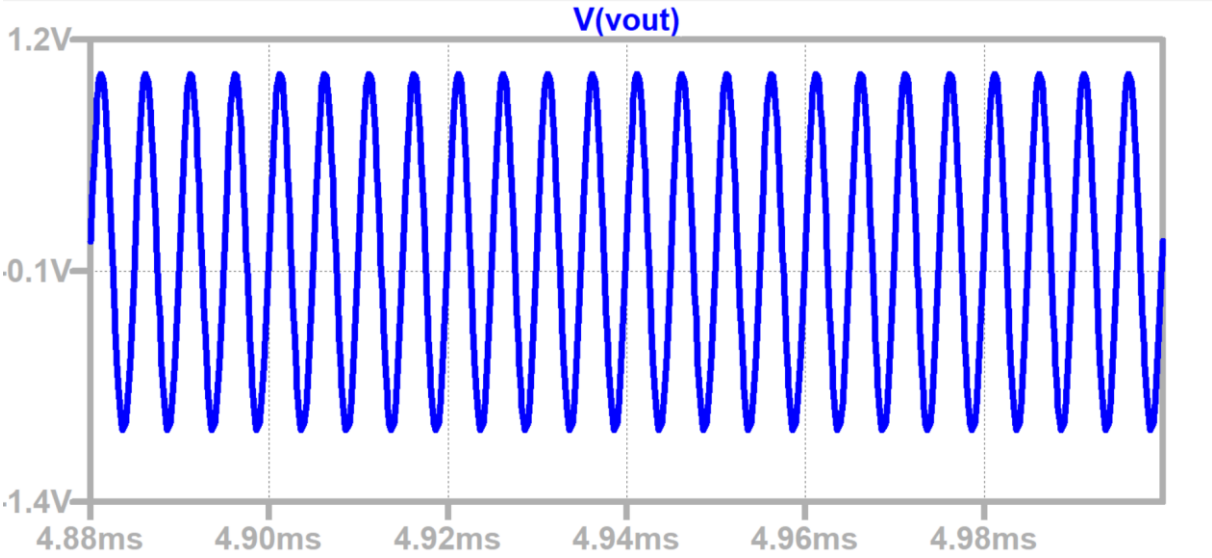

Figure 2: Vout for Input=200kHz and 0.01V

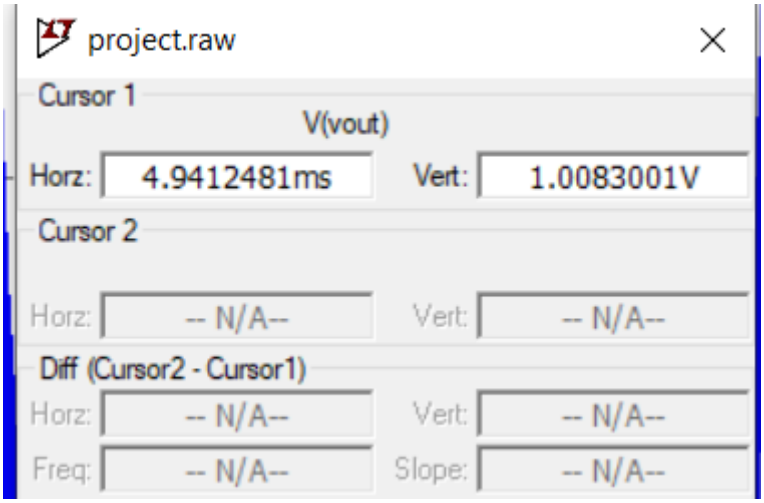

Figure 3: Value of Vout

Input=200kHz and 0.1V

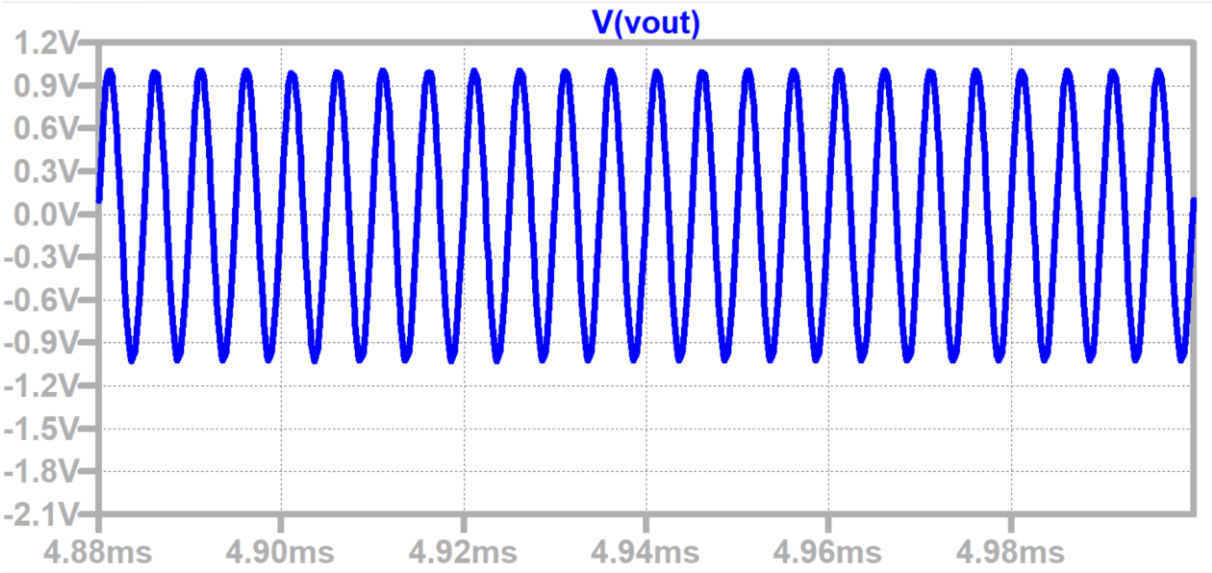

Figure 4: Vout for Input=200kHz and 0.1V

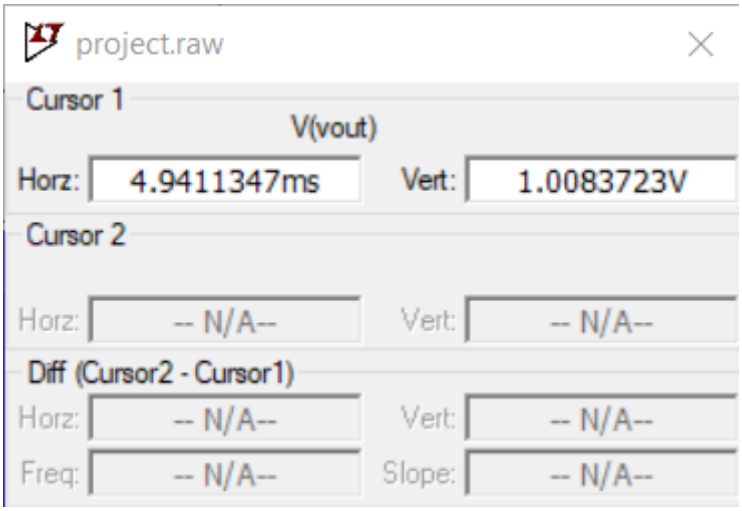

Figure 5: Vout Value

Input=2megaHz and 0.01V

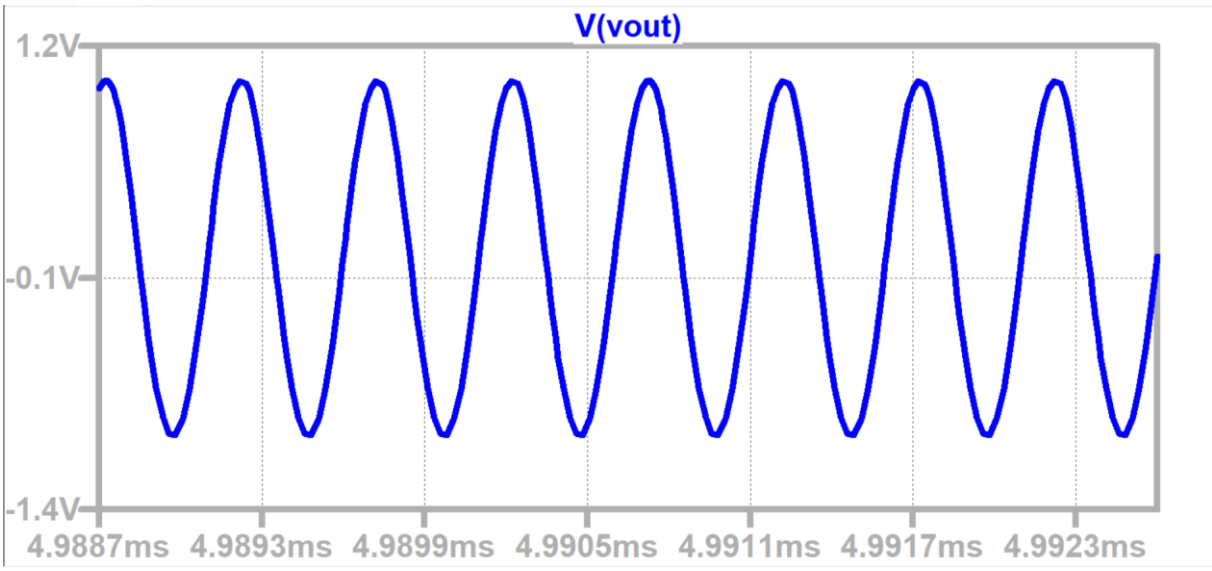

Figure 6: Vout for Input=2megaHz and 0.01V

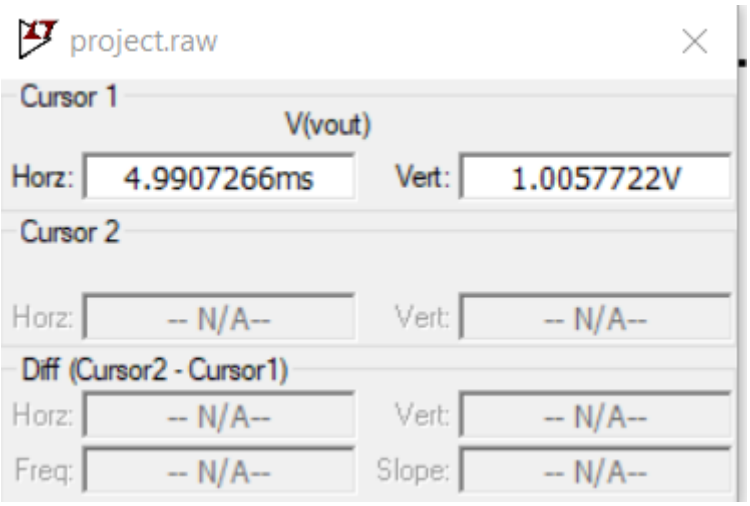

Figure 7: Vout value

Input=2megaHz and 0.1V

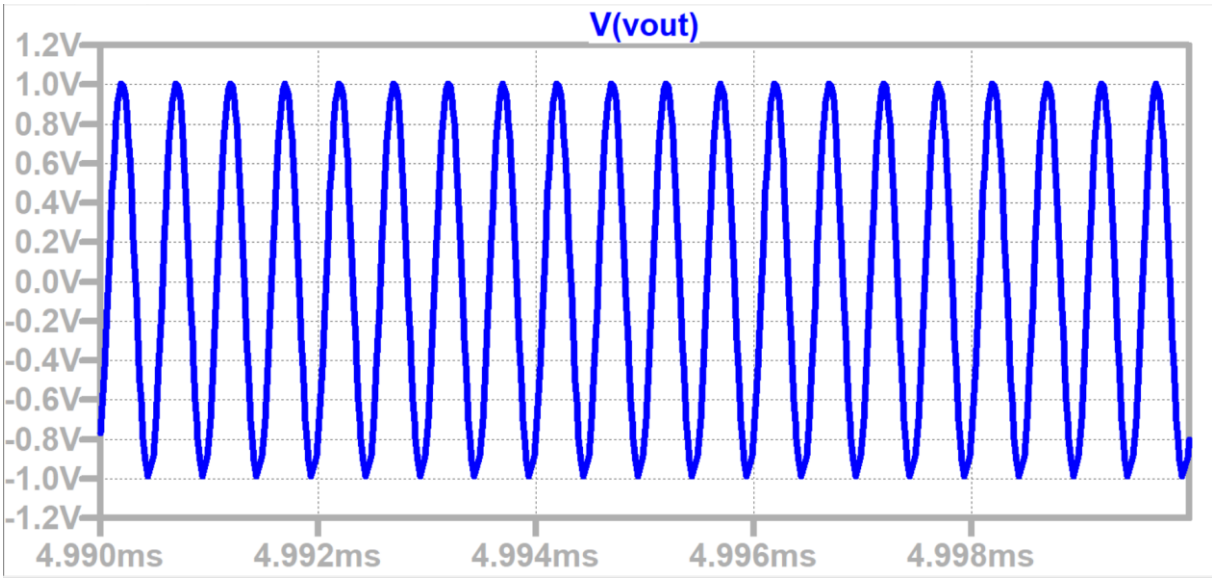

Figure 8: Vout for Input=2megaHz and 0.1V

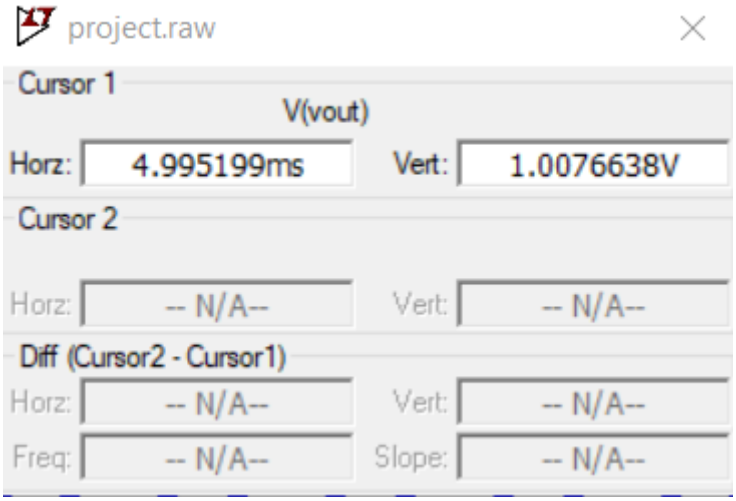

Figure 9: Vout value

As seen, Vout is between 0.9V and 1.1V, so it holds.

Specification 2:

The power supply current, ICC, is less than 100mA.

Input=200kHz and 0.1V

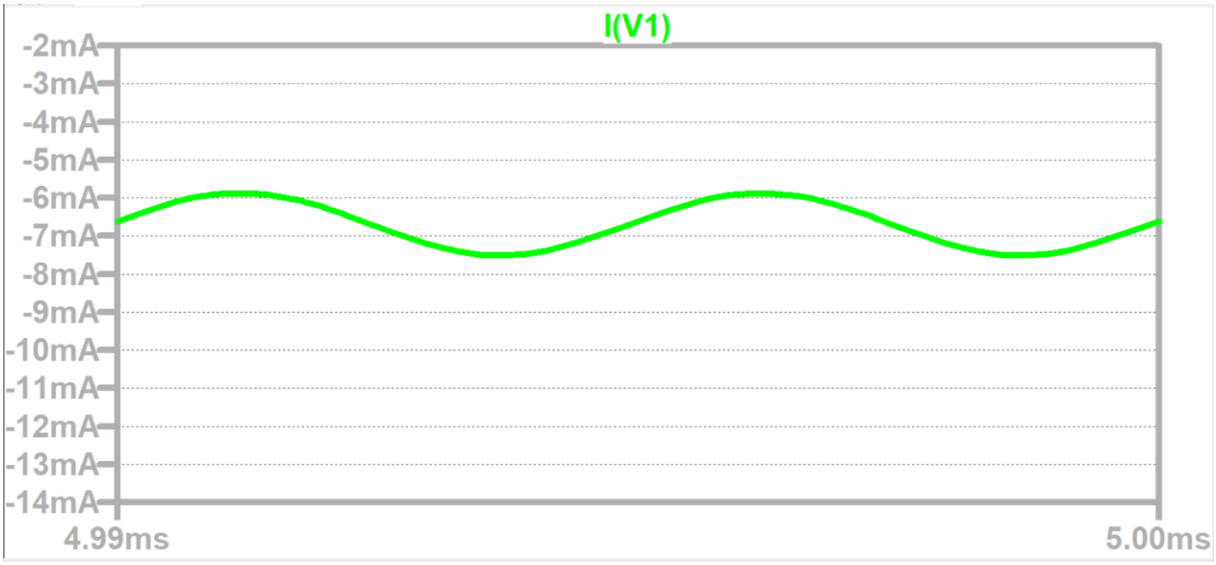

Figure 10: Icc for Input=200kHz and 0.1V

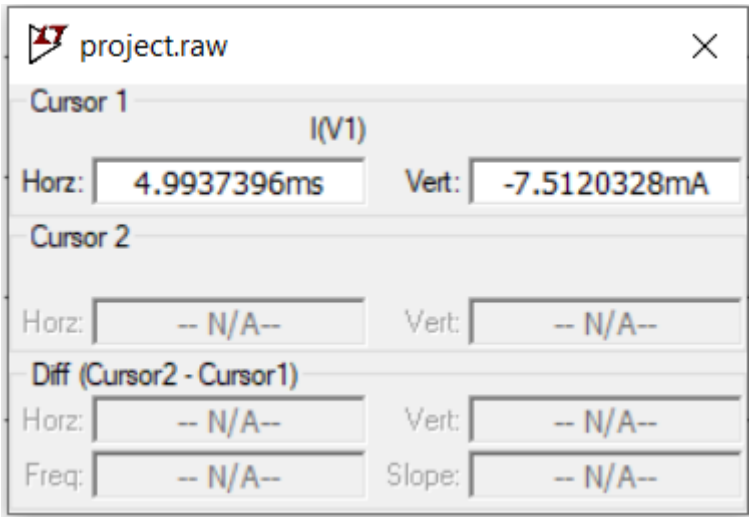

Figure 11: Icc value

Input=200kHz and 0.01V

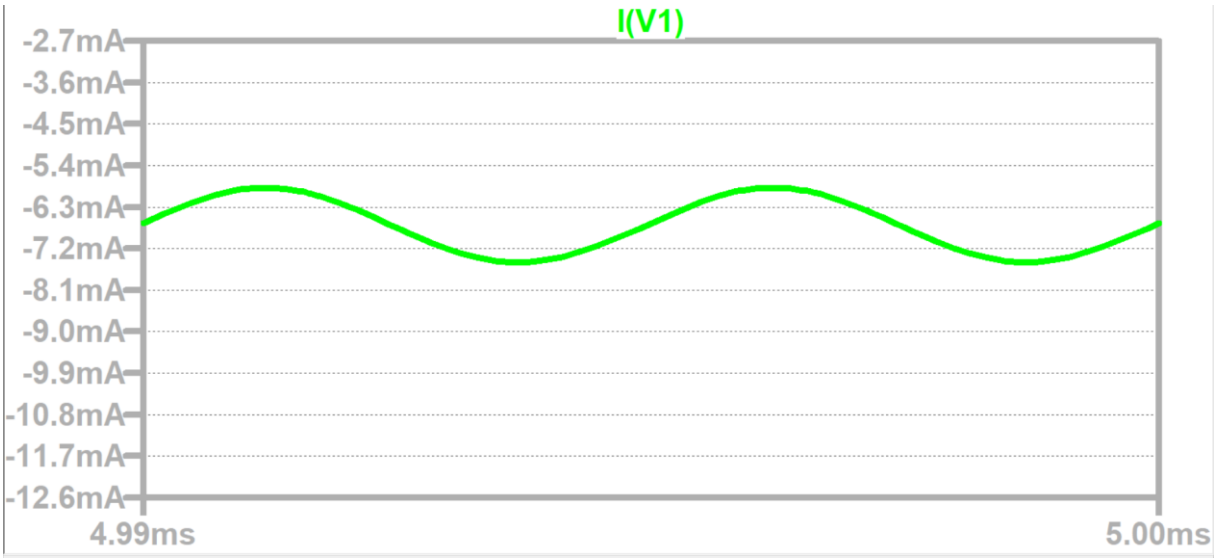

Figure 12: Icc for Input=200kHz and 0.01V

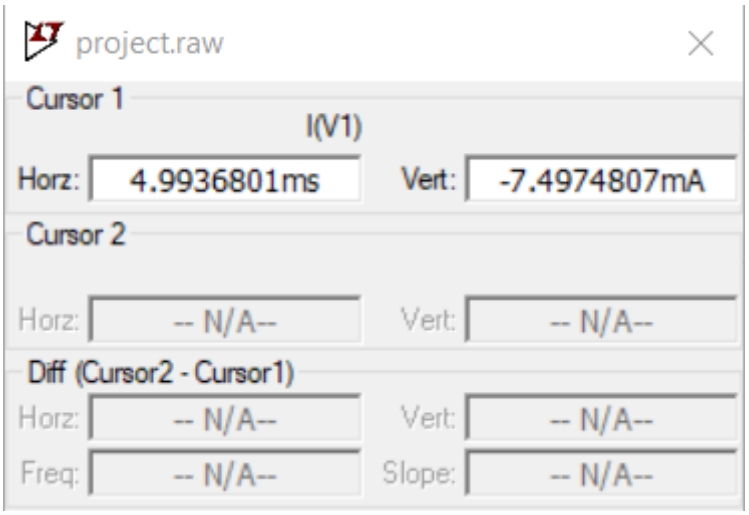

Figure 13: Icc value

Input=2megaHz and 0.1V

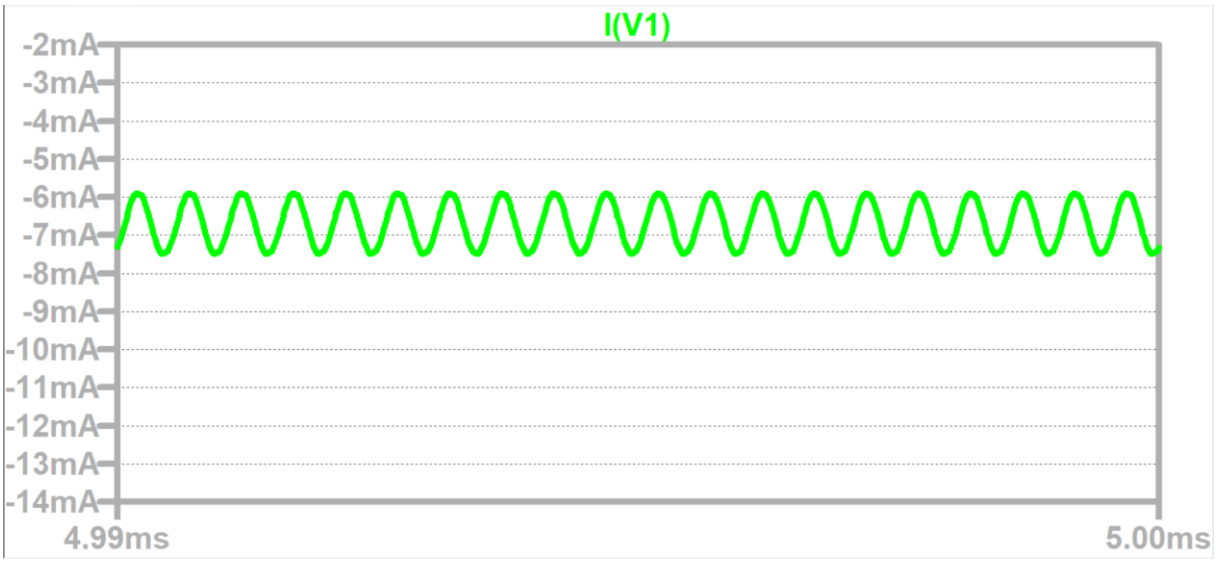

Figure 14: Icc for Input=2megaHz and 0.1V

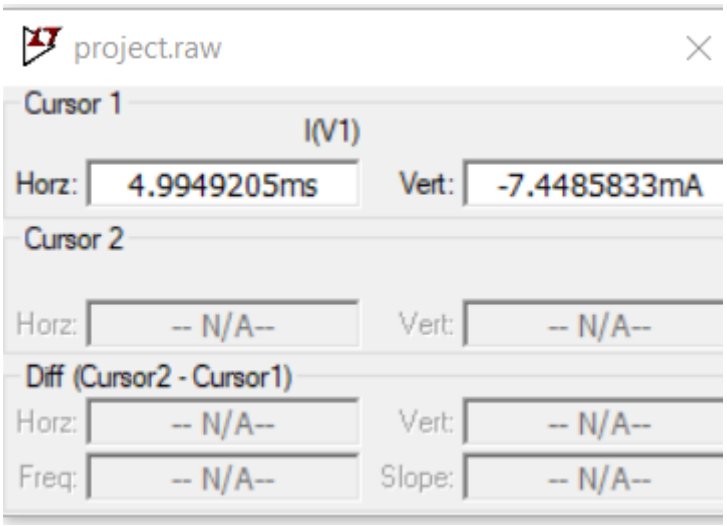

Figure 15: Icc value

Input=2megaHz and 0.01V

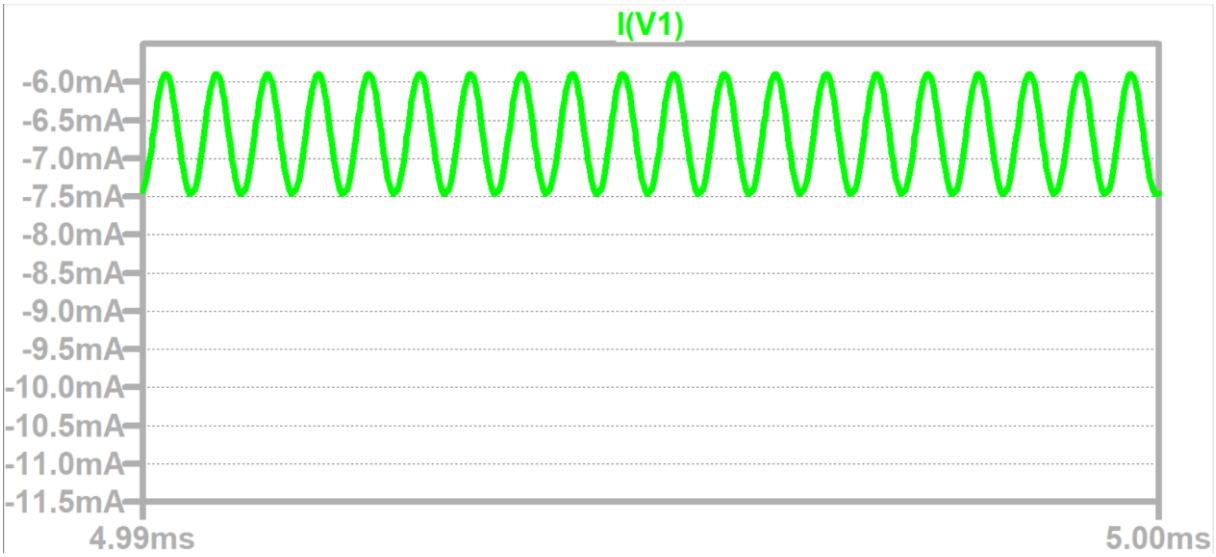

Figure 16: Icc value for Input=2megaHz and 0.01V

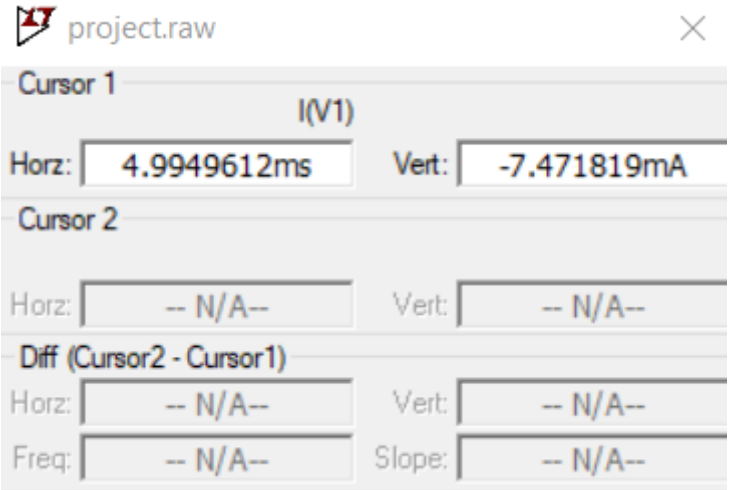

Figure 17: Icc value

As seen Icc is less then 100 mA so it holds.

Specification 3:

LED turns ON when the peak output voltage is Voutp=1±0.1 V. It should turn off if the peak output is less than Voutp< 0.9 V.

Vout is in range and Led is ON

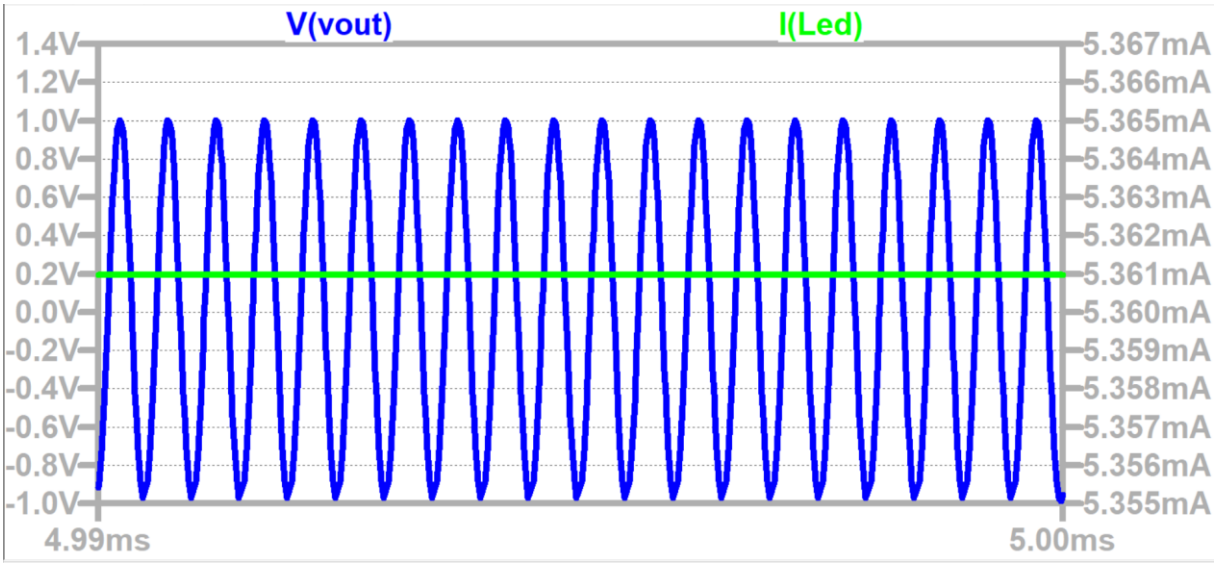

Figure 18: Vout in range and LED is ON

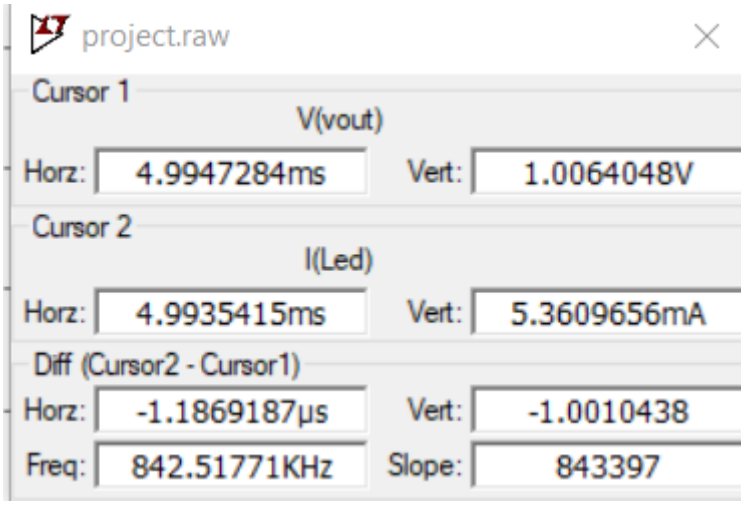

Figure 19: Values of Vout and LED

Vout is not in range and Led is OFF

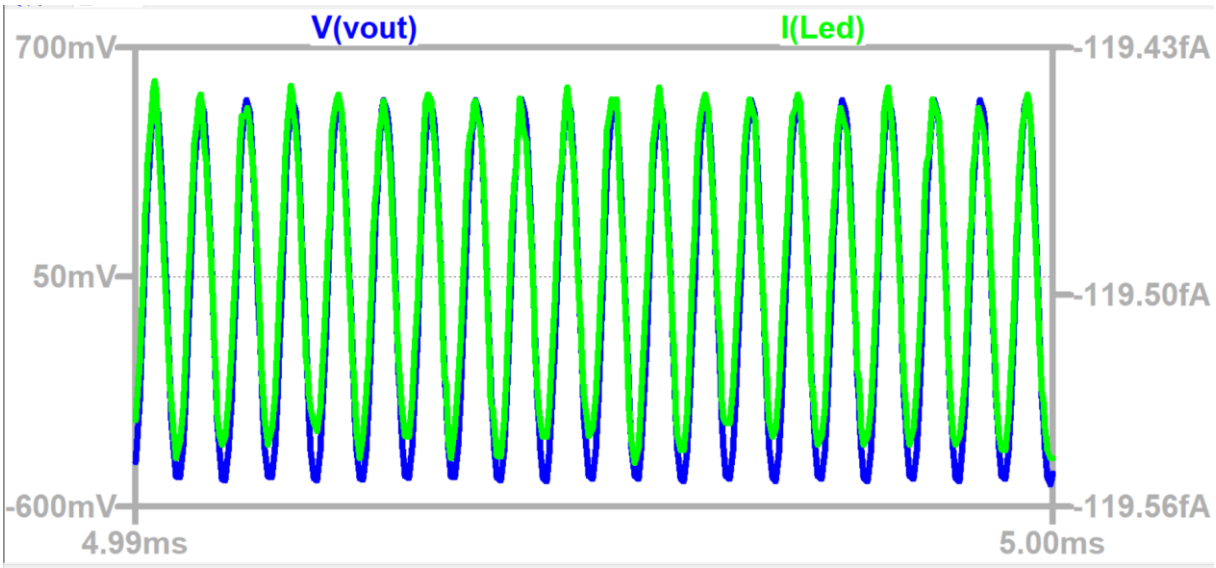

Figure 20: Vout is not in range and Led is OFF

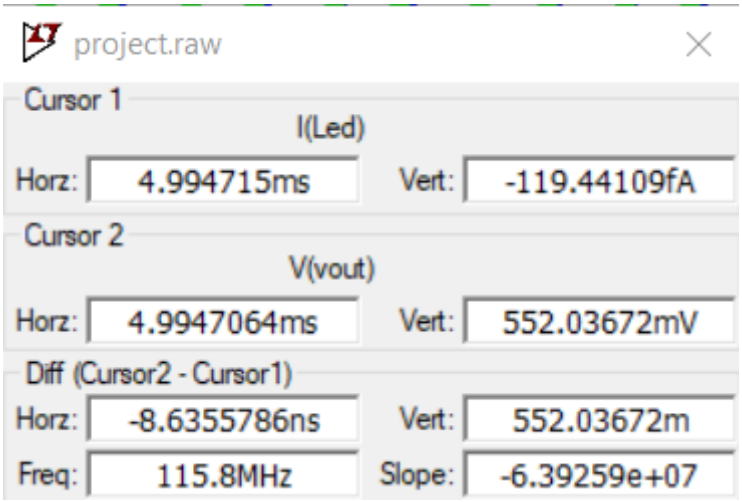

Figure 21: Values for Vout and LED

Specification holds.

TABLE

	Vinp(mV)	Voutp(V)	LED ON/OFF	Icc(mA)
0.2MHz	10	**1.008**	**ON**	**7.5**
0.2MHz	Vinpmin=6	1+-0.1	ON	**7.5**
0.2MHz	100	**1.008**	**ON**	**7.5**
0.2MHz	Vinpmax=400	1+-0.1	ON	**7.4**
2MHz	10	**1.006**	**ON**	**7.5**
2MHz	Vinpmin=10	1+-0.1	ON	**7.5**
2MHz	100	**1.008**	**ON**	**7.4**
2MHz	Vinpmax=600	1+-0.1	ON	**7.5**